



Society of Petroleum Engineers

SPE-227354-MS

Probing the Generation of Reactive Species from Surfactants with Interface-Sensitive Mass Spectrometry Studies

Theis I. Sølling, Safwat Abdel-Azeim, Ahmed M. El Zohry, and Syed Muhammad Shakil Hussain, Center for Integrative Petroleum Research, College of Petroleum & Geosciences, King Fahd University of Petroleum & Minerals, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227354-MS](https://doi.org/10.2118/227354-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

We report the first observation of a hypervalent nitrogen species stabilized in the condensed phase with a lifetime on the order of minutes, formed via homolytic N–Br bond cleavage in a gemini surfactant at the air-liquid interface. The interfacial environment, shaped by entangled hydrophobic-hydrophilic surfactant structures resembling porous membranes, promotes radical formation over ionic dissociation. Simultaneously, we explore the unique interfacial behavior of Zn^{2+} compared to other transition metals using Field-Induced Droplet Ionization (FIDI). Zn^{2+} alone exhibits distinct speciation at the interface, forming a stable complex with three ethanol ligands—a behavior attributed to its electronic configuration and optimal polarity. Together, these findings highlight how interfacial architecture and specific ion properties can stabilize otherwise transient species and open pathways for designing controlled-release systems for reactive intermediates.

Introduction

The characterization methods of reactions and structure at interfaces are few and far in between. This contribution will focus on applications of interface-sensitive mass spectrometry in the oil and gas sector and also to some extent beyond. We have published three papers in the peer-reviewed literature on the topic recently. (Sølling et al., 2025; T. I. Sølling, Andersson, et al., 2024; T. I. Sølling, El Zohry, et al., 2024) Firstly, one, which is currently under review particularly shows how highly reactive species can be formed under sets of conditions that are dictated by the interfacial structure. It will form most of the basis for this conference contribution. Secondly, we have also looked at how electrolytes speciate at interfaces. (T. I. Sølling, Andersson, et al., 2024) This is an important aspect in for example emulsification challenges. Thirdly, various tools were investigated keeping the crude oil application in mind. (T. I. Sølling, El Zohry, et al., 2024)

Interfacial phenomena play a critical role in surfactant applications across various fields. Surfactants, or surface-active agents, function by positioning themselves between two immiscible phases, reducing the molecular tension caused by the repulsion between these incompatible substances. (Binks, 2002;

Prosser & Franses, 2001; Yin & Alivisatos, 2005) The behavior of the interfacial layer, along with how surfactants distribute between the bulk phases and the interface, directly influences their performance in different applications.(Masoudi et al., 2019; SCRIVEN & STERNLING, 1960) A key aspect in all cases is determining whether surfactants actually localize at the interface, and if so, whether they undergo chemical reactions there.(Masoudi et al., 2019)

When surfactants are stably present at an interface, they typically reduce interfacial tension. This is especially beneficial in processes like enhanced oil recovery (EOR), where lowering the tension between oil and water helps emulsified fluids move from small capillaries into larger channels and eventually into the production well. In contrast, for applications such as CO₂ storage, the goal is to inject and retain the supercritical fluid in tiny pore spaces.(Hirasaki et al., 2011; Jia et al., 2019)

Optimizing surfactant structures for EOR involves several design factors, with a primary focus on improving solubility and thermal stability in high-salinity brine conditions.(S. M. S. Hussain et al., 2019) However, limited attention has been given to the specific reactivity patterns that might occur at interfaces. Our previous work has examined the composition of air-crude oil interfaces and interfaces between air and aqueous solutions containing transition metals, using a surface-specific technique known as Field-Induced Droplet Ionization (FIDI) mass spectrometry.(T. I. Solling, Andersson, et al., 2024) These studies provided vital insights into the interfacial composition of crude oil and brines.

In this study, we focus on the interfacial reactivity of diammonium groups—the core structure of gemini surfactants commonly used in oil and gas applications. These surfactants are engineered to endure the harsh conditions of high salinity and temperature found in subsurface environments, and to resist degradation in brines—a longstanding challenge in the industry(S. Hussain et al., 2019). Previous findings have shown that in the gas phase, ammonium ions can capture an extra electron, forming a hypervalent nitrogen species with nine electrons—though this behavior has not been observed in condensed phases. Here, we investigate how this unique structural feature affects the interfacial behavior of diammonium-based gemini surfactants and whether distinct reactivity mechanisms emerge at the interface.

Experimental Description

Electrospray Ionization (ESI) Mass Spectrometry

To characterize the molecular composition of molecules in general, electrospray ionization (ESI) mass spectrometry is frequently employed. The ESI spectra herein were acquired using a Bruker MicroTOF-III mass spectrometer, following the standard operating procedure adopted for the characterization of synthetic products within our department. This instrument is routinely used for high-resolution mass spectral analysis and is well-suited to the profiling of synthetic surfactants due to its precision and reproducibility. Essentially the way that it is done is by forming the solutions and using a Hamilton syringe to inject the solution into the mass spectrometer at a rate which gives sufficient signal.(S. M. S. Hussain et al., 2019)

In addition to the standard ESI analysis, more advanced and custom-designed mass spectrometric experiments were conducted using a ThermoFisher LTQ mass spectrometer. These experiments utilized a custom-built ionization source specifically tailored for Field-Induced Droplet Ionization (FIDI), an emerging surface-sensitive ionization technique that offers unique capabilities for probing interfacial chemical phenomena.

Before each ESI measurement, the Bruker mass spectrometer was calibrated using sodium trifluoroacetate (NaCF₃COO) as a calibration standard. This procedure ensured a consistent mass resolution of at least 10⁴, even at high mass-to-charge ratios (up to 1600 u), allowing for the accurate identification of larger oligomeric and polymeric species, such as those commonly observed in surfactant structures.

FIDI calibration, in contrast, involved matching peak patterns to expected sidechain structures, such as differing numbers of ethylene oxide (CH₂CH₂O) and methylene (CH₂) units. This qualitative calibration

method was essential due to the complexity and variability in sidechain composition inherent in the gemini surfactant mixtures.

Given that this surfactant is a regularly synthesized compound in our laboratory, its ESI spectrum has been recorded numerous times and shown to be highly reproducible, offering a reliable reference point for subsequent FIDI measurements. FIDI experiments were typically conducted multiple times per day, often as many as five separate acquisitions, and spread across at least three non-consecutive days over the course of a month-long data acquisition campaign. This ensured robust sampling and high statistical confidence in the spectral data.

Field-Induced Droplet Ionization (FIDI)

Field-Induced Droplet Ionization is a surface-sensitive mass spectrometry technique that has been previously described in detail in earlier publications.(Grimm & Beauchamp, 2003) In this study, the "hanging droplet" mode was employed, where a droplet of the analyte solution is suspended from a capillary tip and subjected to a high-voltage pulsed electric field to induce ion formation.

One critical factor that strongly influences FIDI performance is the volume of the droplet. It was found through systematic variation and optimization that a droplet volume of 33 microliters, corresponding to a diameter of approximately 2–3 millimeters, provided the most stable and intense ion signals. The sensitivity of the FIDI technique to droplet volume arises from its influence on the formation of opposing Taylor cones, which are responsible for ejecting analyte ions toward the detector.

To create these Taylor cones and generate ionization, a pulsed voltage of approximately 5 kilovolts was applied. For a droplet of 33 μL composed of a hydrocarbon solvent, this voltage corresponds to a field strength of approximately 2.5 kV/cm. This value was determined iteratively, with ion peak intensity and reproducibility across pulses serving as key performance metrics.(Grimm & Beauchamp, 2003)

Synthesis of Gemini Surfactants

The gemini surfactants analyzed in this study were synthesized in-house using a method previously published and validated in the literature. They have been shown to be thermally stable, resilient to high salinities and also soluble in highly saline brines. Coreflooding experiments have shown that they enhance the recovery (in core flooding runs) with up to 20 percent. The synthetic process results in a product containing a distribution of ethoxy units in the side chains. These ethoxy variations carry over from the starting material to the final product, giving rise to a structurally heterogeneous surfactant mixture. Consistent with industry practice, the product was used without further purification, as separating ethoxy chains by specific length is both technically challenging and economically impractical.

The surfactant was subsequently dissolved in a variety of solvents, including water, methanol, ethanol, a 20:80 methanol–methylene chloride mixture, and crude oil in toluene, to achieve concentrations of approximately 1000 ppm. At this concentration, multilayer surface coverage is expected (see Supporting Information), which is suitable for studying interfacial phenomena.(S. Hussain et al., 2019)

This multi-solvent approach allowed for systematic exploration of how solvent polarity and hydrogen bonding interactions influence the diffusion behavior and chemical speciation at the interface. Importantly, these experiments were conducted prior to introducing crude oil, which presents a far more chemically complex and variable medium.

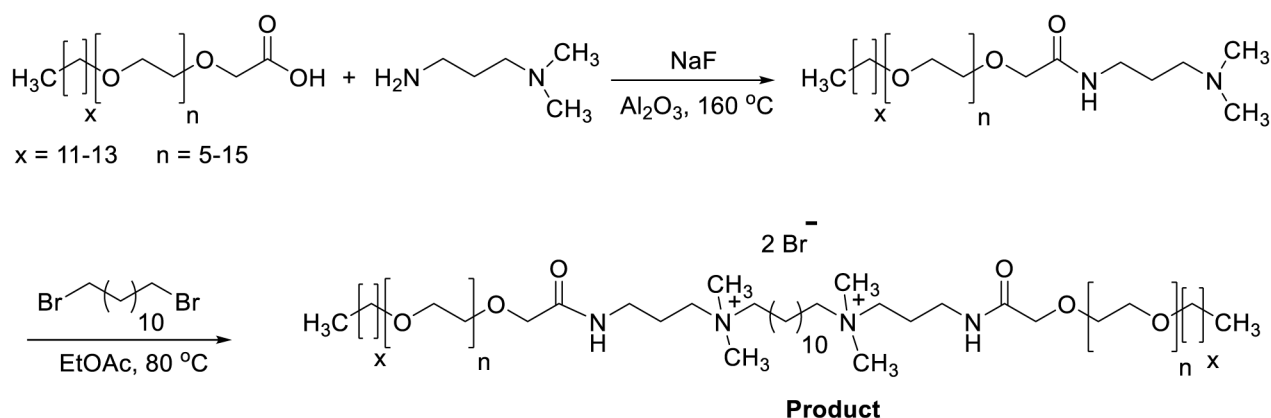


Figure 1—(adapted from (Solling et al., 2025)) outlines the synthetic route and molecular structure of the surfactant under investigation. This particular compound was chosen due to its high thermal stability and resilience in high-salinity environments, properties that make it highly attractive for applications in geothermal energy systems and oil/gas reservoirs. These environments require surfactants that can remain functional under extreme temperature and ionic strength conditions.

Due to the inherent compositional diversity of the ethoxy-containing sidechains, the resulting surfactant mixture lacks uniformity, leading to a disordered (amorphous) interfacial environment. This mismatch in chain length among the surfactant molecules contributes to inefficient molecular packing at the interface. To investigate this phenomenon further, molecular dynamics simulations will be used to examine and verify the amorphous nature of the interfacial layer, even in systems where sidechain lengths are deliberately controlled.

Molecular dynamics simulations

Atomistic-level structural information of the gemini surfactant at the air–water interface was obtained through molecular dynamics (MD) simulations. The interface was modeled using a simulation box measuring $8.2 \times 8.2 \times 40.0 \text{ nm}^3$, providing a surface area of 67.2 nm^2 . Simulations were executed using GROMACS 2022.3 (Hess et al., 2008) with the OPLS-AA force field (Jorgensen et al., 1996; Jorgensen & Tirado-Rives, 1988) and SPCE water model (Berendsen et al., 1987), which have shown good performance in similar systems (Abdel-Azeim & Kanj, 2018; Alade et al., 2020; Bořan et al., 2016; Mafi et al., 2017). Parameters for bromide ions followed the work by Jensen and Jorgensen (2006).

Prior to production runs, the system was energy-minimized using the steepest descent algorithm with a convergence threshold of $100 \text{ kJ mol}^{-1} \text{ nm}^{-1}$. Long-range electrostatics were computed via the Particle-Mesh Ewald (PME) method (Darden et al., 1993), with a 1.2 nm cutoff applied to both van der Waals and short-range Coulombic interactions. Bond constraints involving hydrogen atoms were handled using the LINCS algorithm (Hess et al., 1997).

Temperature control was achieved with the Nosé–Hoover thermostat (Nosé, 1984; Hoover, 1985), using time constants of 0.1 ps during equilibration and 0.5 ps for production. Equilibration was performed in the NPT ensemble, followed by a 100 ns production run under NVT conditions. Simulations used a 2 fs timestep, and atomic coordinates were stored every 10 ps for post-processing. Trajectory analysis was conducted using built-in GROMACS tools, and the initial system configuration was constructed using the PACKMOL software (Martínez et al., 2009).

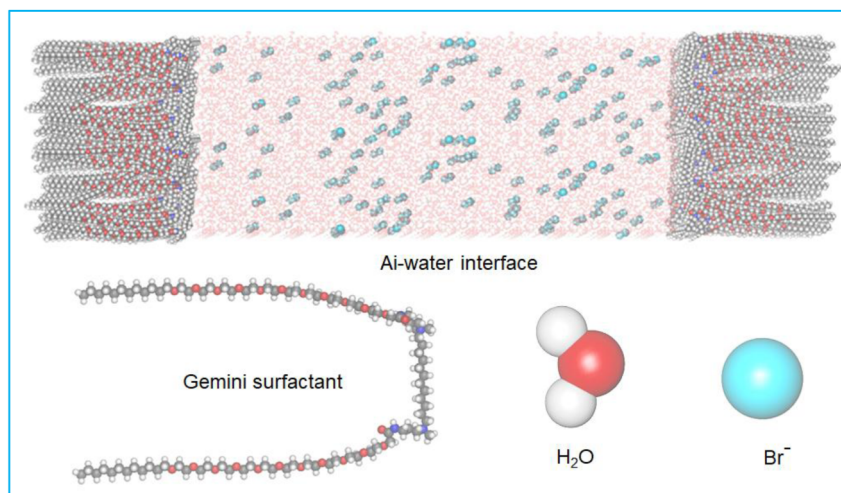
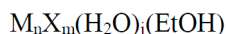


Figure 2—(adapted from Solling et al., 2025) The molecular model used for the simulation of the air-water interface in the presence of gemini surfactant.

Results and Discussion

Measurements to reveal the behavior of metal ions at interfaces

Outcomes of FIDI experiments are of central importance in several topics that are key to the upstream sector. An example is pertaining to speciation of metal ions at the interface between air and brine. (T. I. Solling, Andersson, et al., 2024) We have recently studied several solutions of transition metal ions and have found specific application in the oil and gas sector in the way that a range of cations arrange themselves at an interface. (T. I. Solling, Andersson, et al., 2024) The key results concerning both the bulk solution and the interface reveal that ionic complexes form with different numbers of water and ethanol molecules coordinated to the central metal ion. Using Field-Induced Droplet Ionization (FIDI), the analysis primarily detects singly charged species that reside at the air–liquid interface. This single charge is stabilized by the anions present in the original solution, leading to a consistent pattern: divalent metal ions are typically observed bound to one anion, while trivalent metal ions coordinate with two anions. Collectively, these complexes conform to the general structural formula illustrated in Scheme 1.



Scheme 1—In this formula M is the metal ion, X is the anion and n, m, j, and l are integers.

In this case, the following findings were made, we particularly notice that Zn^{2+} behaves as a surfactant in many cases: Overall, while some spectral peaks appear in both bulk and interfacial measurements, the ESI and FIDI spectra show marked differences for all examined divalent cations. Typically, FIDI spectra exhibit a greater presence of high molecular weight complexes, as exemplified by the Fe(II) data. A notable exception is Zn^{2+} , where the FIDI spectrum is dominated by a single major species. The most intense peaks in the ESI spectra consistently involve complexes with two ethanol ligands, whereas those in the FIDI spectra involve three ethanol ligands. Most complexes adopt a tetrahedral geometry, with the exception of Cu in the ESI data, which is coordinated by eight ligands. This suggests a general trend where the number of ethanol ligands differs between bulk and interface, but remains consistent across the metal ions within each phase.

Experimental observations point to a high degree of selectivity in complex formation, though significant divergence between ESI and FIDI results is only clearly observed for Zn^{2+} . Broadly, the coordination environments that are energetically favored include two or three ethanol ligands. In the bulk phase, as reflected by ESI data, complexes with two or three ethanol molecules are predominant, indicating variable

hydration levels. These trends are further supported by theoretical calculations, which predict that such coordination environments are the most stable in bulk. (T. I. Solling, Andersson, et al., 2024)

At the interface, the polarity of the surrounding environment plays a key role. The air interface, being non-polar, tends to stabilize less polar complexes—favoring those with more ethanol ligands and thus more hydrocarbon content. This explains why three-ethanol complexes are more commonly observed at the interface, in contrast to the bulk where two-ethanol species dominate. The polar nature of the bulk phase likely drives this difference. Apart from the ligand count, the primary observation is that all metal ions, with the exception of Zn^{2+} , exhibit diverse and non-specific coordination patterns at the interface, despite the general predominance of three-ethanol complexes.

Zn^{2+} , however, shows a strikingly simple FIDI spectrum, with $[\text{ZnX}(\text{EtOH})_3]^+$ being essentially the only detected species. Interestingly, this pronounced specificity is not mirrored in the computational results, which do not predict $[\text{ZnX}(\text{EtOH})_3]^+$ to be especially favored energetically. This discrepancy may suggest that the observed preference arises from effects not fully captured in the calculations or lies within their inherent error margins. Given that equilibrium constants are exponentially dependent on energy differences, even minor variations can lead to significant differences in observed species. The following section explores potential reasons behind Zn^{2+} 's unique behavior.

The interfacial characteristics of transition metal brines in contact with air generally exhibit consistent trends in terms of ligand coordination. At the interface, metal ions are most commonly surrounded by three ethanol ligands. This preference is largely driven by the non-polar nature of the air phase and the tendency for polarity to decrease as more hydrocarbon chains (from ethanol) are attached to the metal center. A particularly striking case is Zn^{2+} , which forms only one dominant complex at the interface. This behavior is attributed to Zn^{2+} 's electronic structure, specifically coordination involving its 4s orbital. In contrast, the bulk phase does not exhibit such selectivity. Earlier research has suggested that Zn^{2+} displays unusually strong and specific interfacial behavior, and the current experimental findings provide direct evidence to support that claim. (Binabdi et al., 2023; T. Solling et al., 2020)

Measurements to reveal the unique behavior of entangled surfactant tails at interfaces: formation of hypervalent species

We present a summary of mass spectrometry and computational findings focused on the behavior of a gemini cationic (diammonium) surfactant in both the bulk (solvent) and at the air-solvent interface. The field-induced droplet ionization (FIDI) method applies a strong electric field to form a Taylor cone, selectively sampling the outermost molecular layers of a hanging droplet—capturing species specific to the interface. In contrast, conventional electrospray ionization (ESI) disperses droplets into an aerosol, reflecting bulk-phase composition. The combined data highlight a unique interfacial reaction pathway in which the surfactant forms a hypervalent nitrogen species, characterized by a nitrogen atom bearing nine electrons.

Bulk-phase interrogations of gemini surfactant mixtures. The dication shown in Figure 1, with parameters $x = 12$, $n = 9$, and $s = 10$, has a molecular weight of 1617.3 g/mol based on its most common isotopes. However, the recorded spectra show no signal at this m/z value or at half that value, which would correspond to a doubly charged ion. Instead, the dominant peak appears at m/z 724.52 u. The spectrum reveals four distinct clusters of peaks, each spaced by 44 u—consistent with variations in the number of ethoxy ($\text{CH}_2\text{CH}_2\text{O}$) units. These clusters are visually represented by a reference peak in Figure 3 and further analyzed with a fit in Figure 4. Notably, there is no evidence of bromine-containing species, as indicated by the absence of characteristic isotopic patterns.

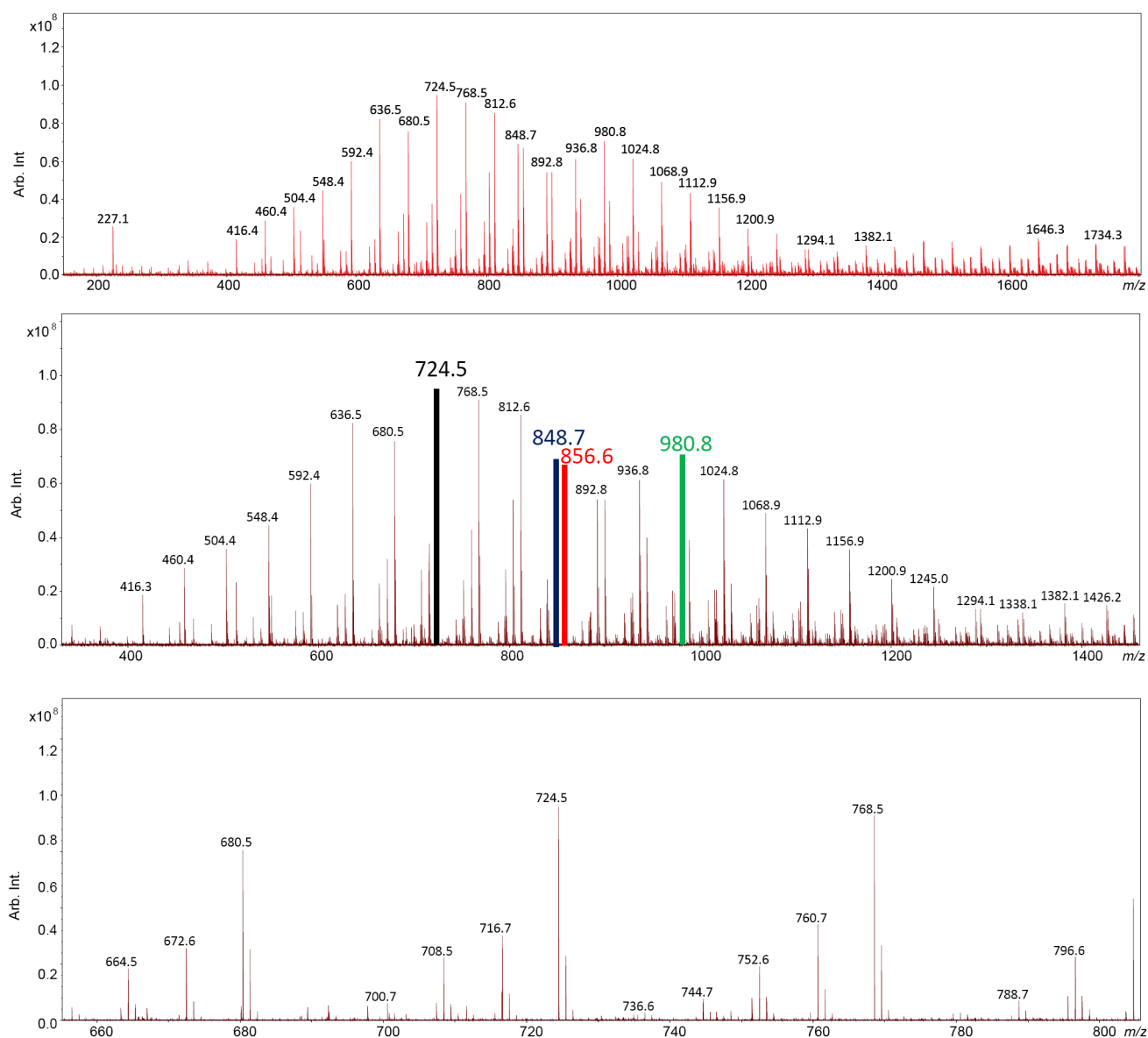
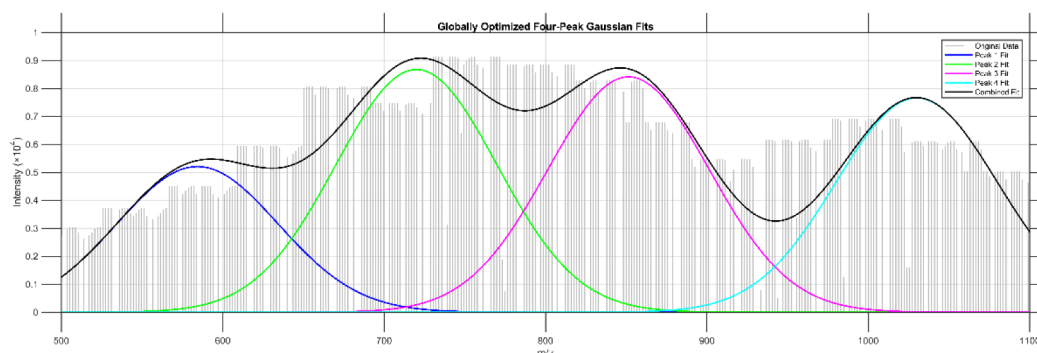
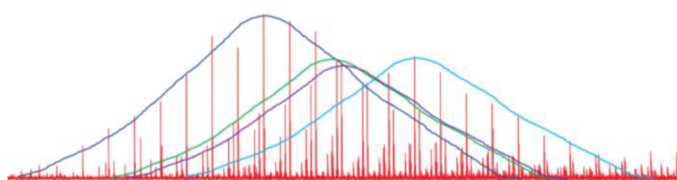


Figure 3—Three different ranges of the electrospray ionization spectrum (in water) of the gemini cation surfactant. Top m/z 0 – 1800, middle 0 – 1450 u, and bottom 550 – 850 u. The range in the middle shows all 4 distributions with a color applied to the central peak to guide the eye. Adapted from (Solling et al., 2025)



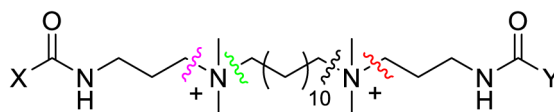
(a)



(b)

Figure 4—(a) Matlab fit of an ESI data subset, the best fit is achieved with 4 gaussian distributions. The sum of center values of the green and the magenta distributions is roughly equal to the sum of the blue and the turquoise. The resulting value is within the range of the average molar weight of the gemini dicationic surfactant. (b) Graphic illustration of the distributions based on the base peaks in Figure 3 (drawn by hand). Adapted from (Solling et al., 2025)

There is no significant peak contribution in the m or $m/2$ region (e.g. at 808.5). This indicates that the dication almost fully decomposes. The observance of four distributions shows that the decomposition takes place via 4 different channels:



Field-Induced Droplet Ionization (FIDI). We investigated the speciation at the air-liquid interface using FIDI, analyzing spectra at two time points: immediately after droplet formation and approximately one minute later. As illustrated in Figure 5, experiments were performed in water to represent a highly polar, hydrogen-bonding environment. Initially, the spectra displayed two ion distributions, with a strong signal from high m/z species corresponding to the gemini surfactant. After a 60-second delay (manually timed), this high-mass distribution disappeared, leaving a single set of lower m/z ions. This time-dependent transformation indicates a clear interfacial decomposition process. In water, the shift was complete within the observation window, showing that the interfacial chemistry proceeds efficiently under aqueous conditions. The initial intensity ratio between the ions at m/z 1573 and 988 was 2.3, further supporting the observed decay behavior.

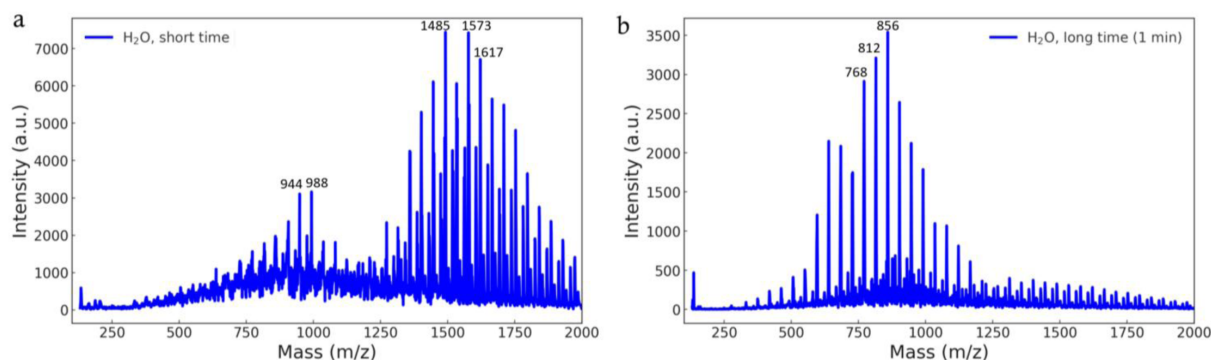


Figure 5—The results of field-induced droplet ionization in various solvents immediately after dispensing the droplet (left) and after waiting approximately 60 sec. (right).

Analysis of the spectra yields several notable insights, the most significant being related to the identity of ions around 1600 u. Given that the synthesized molecule shown in Figure 1 carries two positive charges, its signal would be expected near $m/z \sim 800$. A singly charged variant might be considered if a bromine atom remains bound to the gemini structure. However, the isotopic distribution in the observed peaks rules out this possibility—there is no evidence of the distinctive bromine isotope signature, which typically shows a 50.7:49.3 ratio for the 79 and 81 u isotopes, resulting in a characteristic $m+2$ pattern. The absence of this pattern indicates that bromine is not present in the detected species. An alternative explanation for the singly charged ion involves electron addition to one of the quaternary ammonium groups, forming a hypervalent nitrogen center with a nine-electron configuration.

To investigate the unusual stability and formation of hypervalent nitrogen species at the air-water interface, molecular dynamics (MD) simulations were used to examine the structural environment of gemini surfactants. A key factor in this process is solvation—effective heterolysis of the bromide salt requires sufficient solvent to accommodate and stabilize both bromide ions and dications, which is especially critical in systems with strong electrostatic repulsion.

At the interface, this condition is not met. The alternating polar and nonpolar nature of the surfactant side chains leads to a disordered, entangled structure resembling an amorphous membrane. This arrangement creates voids that allow partial, but not complete, solvation of bromide ions, limiting their mobility and preventing full ionic separation. Instead of maintaining a stable ion-pair configuration, the system appears to favor charge neutralization via bromine atom release—driven by repulsive interactions between the two positive charges on the diammonium core.

Further computational analysis supports this mechanism. The spatial distribution of bromine atoms shows a strong tendency for close Br–Br proximity without intervening solvent molecules. The typical separation distance between these atoms is consistent with van der Waals contact but insufficient to accommodate a water molecule, suggesting a high likelihood of Br₂-like interactions forming under these constrained interfacial conditions. These results reinforce the hypothesis that steric and solvation limitations at the interface promote bromine atom release and the transient formation of a hypervalent nitrogen species.

Interfacial Chemistry. Molecular dynamics simulations reveal significant structural disorder at the air-water interface, which is further intensified by the diversity in surfactant tail structures. The resulting interfacial layer resembles a porous, three-dimensional network where the branched hydrocarbon chains are dispersed irregularly. This disordered architecture creates a confined environment for the gemini surfactant's ionic components, limiting solvation and restricting the charge separation typically observed in aqueous solutions of diammonium salts. The constrained space and reduced water content inhibit traditional solubilization behavior.

This limited solvation environment explains why changes in solvent polarity have minimal impact on FIDI spectra—indicating that specific solvation does not govern the observed chemistry. Instead, under

these unique interfacial conditions, the diammonium ion undergoes N–Br homolysis, releasing a neutral bromine atom and forming a hypervalent nitrogen center. The system is energetically driven toward generating a distonic ion, where the charge and radical sites are spatially separated. A sequence of hydrogen shifts ultimately leads to the loss of a stable alkene and formation of an amine radical cation. Computational studies show this pathway to be highly exothermic, yet the interface's entangled morphology retards these rearrangements, stabilizing the hypervalent intermediate long enough for experimental detection.

While bromine's gas-phase electron affinity is well known (324.6 kJ/mol), this alone doesn't capture the full energetic context of bromide-to-bromine conversion here, which is influenced by solvation energies, electrostatic repulsion within the dication, and electron placement on the ammonium nitrogen. Crucially, the time-dependent nature of the signal confirms that the hypervalent species is not an artifact of the ionization process itself.

Hypervalent species, while not new,^(MUSHER, 1969; REED & SCHLEYER, 1990; Zhdankin, 2016) are typically associated with heavier elements such as iodine.^(Leusser et al., 2004) Reports of second-period atoms exceeding the octet rule exist but often involve highly controlled or theoretical conditions. Although gas-phase studies, like those by Tureček, have documented hypervalent nitrogen,^(Frosig & Turecek, 1998; Holm et al., 2008; Yao & Turecek, 2005) condensed-phase examples remain rare. The mechanism presented here, driven by interfacial confinement and charge frustration, is both uncommon and difficult to study due to the limited availability of surface-specific characterization methods such as FIDI.

Other studies have similarly highlighted the unique reactivity at interfaces. For instance, Cohen et al. demonstrated enhanced chlorination of double bonds due to molecular orientation at air interfaces, while the Zare group has observed distinct reactivity at water microbubble surfaces.^(Li et al., 2021; Zhou et al., 2025) These findings support the idea that air-liquid interfaces can foster unique chemical behaviors. In this context, the formation of long-lived hypervalent radicals is not just novel—it fits within a broader framework of interface-enabled reactivity documented in the literature.

Conclusion

This study demonstrates the first observation of a condensed-phase hypervalent nitrogen species centered on a second-period atom, stabilized by a structured interfacial environment formed through the entanglement of amphiphilic surfactant tails. This porous, membrane-like interface promotes the homolytic cleavage of N–Br bonds and facilitates the transient stabilization of highly reactive nitrogen species, observable via FIDI. In parallel, we reveal that Zn^{2+} exhibits unique interfacial speciation, forming a well-defined complex with three ethanol molecules due to its specific electronic configuration and polarity optimization at the air-liquid boundary. These findings highlight the critical role of interfacial architecture and coordination chemistry in stabilizing reactive intermediates and tuning interfacial behavior.

Looking ahead, these insights hold particular promise for the upstream oil and gas sector, where controlling interfacial processes is essential—for instance, in enhanced oil recovery, scale inhibition, and emulsion stability. The ability to design interfaces that enable the controlled release of reactive species or fine-tune ion-surfactant interactions opens new possibilities for improving chemical performance in harsh reservoir environments. Further investigations, including advanced techniques such as sum frequency generation, will be vital to fully exploit these mechanisms for operational efficiency and innovation in interfacial engineering.

References

- Abdel-Azeim, S., & Kanj, M. Y. (2018). Dynamics, aggregation, and interfacial properties of the partially hydrolyzed polyacrylamide polymer for enhanced oil recovery applications: Insights from molecular dynamics simulations. *Energy & Fuels*, **32**(3), 3335–3343.

- Alade, O. S., Abdel-Azeim, S., Mahmoud, M., Hamdy, M., Al-Shehri, D. A., & Mokheimer, E. (2020). Studies of Interaction between Bitumen and Thermochemical Fluid (TCF): Insights from experiment and Molecular Dynamics Simulations. *Applied Surface Science*, 146942.
- Solling, T. I., Abdelazeim, S., El Zohry A. M., Hussain, S. M, S.; (2025). The involvement of hypervalent nitrogen in the unique interfacial reactivity of gemini cationic surfactants: an interface-sensitive mass spectrometry study. *Journal of Petroleum Science and Engineering*, **149**, 107–113. <https://doi.org/10.1016/j.petrol.2016.10.034>
- Berendsen, H. J. C., Grigera, J. R., & Straatsma, T. P. (1987). The missing term in effective pair potentials. *Journal of Physical Chemistry*, **91**(24), 6269–6271.
- Binabdi, A. A., Solling, T. I., El Zohry, A. M., Ayirala, S., Gmira, A., & Yousef, A. A. (2023). Fundamentals of crystallization at oil-brine interfaces for conformance control and oil spill remediation applications. *GEOENERGY SCIENCE AND ENGINEERING*, **227**, 211930. <https://doi.org/10.1016/j.geoen.2023.211930>
- Binks, B. (2002). Particles as surfactants—Similarities and differences. *CURRENT OPINION IN COLLOID & INTERFACE SCIENCE*, **7**(1–2), 21–41. [https://doi.org/10.1016/S1359-0294\(02\)00008-0](https://doi.org/10.1016/S1359-0294(02)00008-0)
- Bořan, V., Ustach, V., Faller, R., & Leonhard, K. (2016). Direct Phase Equilibrium Simulations of NIPAM Oligomers in Water. *The Journal of Physical Chemistry B*, **120**(13), 3434–3440.
- Darden, T., York, D., & Pedersen, L. (1993). Particle mesh Ewald: An $N \cdot \log(N)$ method for Ewald sums in large systems. *The Journal of Chemical Physics*, **98**(12), 10089–10092.
- Frosig, L., & Turecek, F. (1998). Hypervalent pyrrolidinium radicals by neutralization-reionization mass spectrometry: Metastability and radical leaving group abilities. *JOURNAL OF THE AMERICAN SOCIETY FOR MASS SPECTROMETRY*, **9**(3), 242–254. [https://doi.org/10.1016/S1044-0305\(97\)00249-3](https://doi.org/10.1016/S1044-0305(97)00249-3)
- Grimm, R. L., & Beauchamp, J. L. (2003). Field-induced droplet ionization mass spectrometry. *Journal of Physical Chemistry B*, **107**(51), 14161–14163. <https://doi.org/10.1021/jp037099r>
- Hess, B., Bekker, H., Berendsen, H. J., & Fraaije, J. G. (1997). LINC: A linear constraint solver for molecular simulations. *Journal of Computational Chemistry*, **18**(12), 1463–1472.
- Hess, B., Kutzner, C., Van Der Spoel, D., & Lindahl, E. (2008). GROMACS 4: Algorithms for highly efficient, load-balanced, and scalable molecular simulation. *Journal of Chemical Theory and Computation*, **4**(3), 435–447.
- Hirasaki, G. J., Miller, C. A., & Puerto, M. (2011). Recent Advances in Surfactant EOR. *Spe Journal*, **16**(4), 889–907. <https://doi.org/10.2118/115386-PA>
- Holm, A., Larsen, M., Panja, S., Hvelplund, P., Nielsen, S., Leib, R., Donald, W., Williams, E., Hao, C., & Turecek, F. (2008). Electron capture, femtosecond electron transfer and theory: A study of noncovalent crown ether 1, n-diammonium alkane complexes. *INTERNATIONAL JOURNAL OF MASS SPECTROMETRY*, **276**(2–3), 116–126. <https://doi.org/10.1016/j.ijms.2008.04.021>
- Hoover, W. G. (1985). Canonical dynamics: Equilibrium phase-space distributions. *Physical Review A*, **31**(3), 1695.
- Hussain, S., Kamal, M., & Murtaza, M. (2019). Synthesis of Novel Ethoxylated Quaternary Ammonium Gemini Surfactants for Enhanced Oil Recovery Application. *ENERGIES*, **12**(9). <https://doi.org/10.3390/en12091731>
- Hussain, S. M. S., Kamal, M. S., Solling, T., Murtaza, M., & Fogang, L. T. (2019). Surface and thermal properties of synthesized cationic poly(ethylene oxide) gemini surfactants: The role of the spacer. *Rsc Advances*, **9**(52), 30154–30163. <https://doi.org/10.1039/c9ra06577f>
- Jensen, K. P., & Jorgensen, W. L. (2006). Halide, Ammonium, and Alkali Metal Ion Parameters for Modeling Aqueous Solutions. *Journal of Chemical Theory and Computation*, **2**(6), 1499–1509. <https://doi.org/10.1021/ct600252r>
- Jia, B., Tsau, J., & Barati, R. (2019). A review of the current progress of CO₂ injection EOR and carbon storage in shale oil reservoirs. *FUEL*, **236**, 404–427. <https://doi.org/10.1016/j.fuel.2018.08.103>
- Jorgensen, W. L., Maxwell, D. S., & Tirado-Rives, J. (1996). Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids. *J. Am. Chem. Soc.*, **118**(45), 11225–11236.
- Jorgensen, W. L., & Tirado-Rives, J. (1988). The OPLS [optimized potentials for liquid simulations] potential functions for proteins, energy minimizations for crystals of cyclic peptides and crambin. *Journal of the American Chemical Society*, **110**(6), 1657–1666.
- Leusser, D., Henn, J., Kocher, N., Engels, B., & Stalke, D. (2004). S=N versus S⁺-N⁻: An experimental and theoretical charge density study. *JOURNAL OF THE AMERICAN CHEMICAL SOCIETY*, **126**(6), 1781–1793. <https://doi.org/10.1021/ja038941+>
- Li, J., Liu, C., Chen, H., & Zare, R. N. (2021). Accelerated Oxidation of Organic Sulfides by Microdroplet Chemistry. *JOURNAL OF ORGANIC CHEMISTRY*, **86**(7), 5011–5015. <https://doi.org/10.1021/acs.joc.0c02942>
- Mafi, A., Hu, D., & Chou, K. C. (2017). Complex Formations between Surfactants and Polyelectrolytes of the Same Charge on a Water Surface. *Langmuir*, **33**(32), 7940–7946.
- Martínez, L., Andrade, R., Birgin, E. G., & Martínez, J. M. (2009). PACKMOL: A package for building initial configurations for molecular dynamics simulations. *Journal of Computational Chemistry*, **30**(13), 2157–2164.

- Masoudi, M., Khosravi, M., Rostami, B., & Abolhosseini, P. (2019). Effect of Benard-Marangoni flow on the bypassed oil recovery: Micromodel study. *JOURNAL OF PETROLEUM SCIENCE AND ENGINEERING*, **178**, 1067–1078. <https://doi.org/10.1016/j.petrol.2019.03.064>
- MUSHER, J. (1969). CHEMISTRY OF HYPERVALENT MOLECULES. *ANGEWANDTE CHEMIE-INTERNATIONAL EDITION*, **8**(1), 54+.
- Nosé, S. (1984). A unified formulation of the constant temperature molecular dynamics methods. *The Journal of Chemical Physics*, **81**(1), 511–519.
- Prosser, A., & Franses, E. (2001). Adsorption and surface tension of ionic surfactants at the air-water interface: Review and evaluation of equilibrium models. *COLLOIDS AND SURFACES A-PHYSICOCHEMICAL AND ENGINEERING ASPECTS*, **178**(1–3), 1–40. [https://doi.org/10.1016/S0927-7757\(00\)00706-8](https://doi.org/10.1016/S0927-7757(00)00706-8)
- REED, A., & SCHLEYER, P. (1990). CHEMICAL BONDING IN HYPERVALENT MOLECULES - THE DOMINANCE OF IONIC BONDING AND NEGATIVE HYPERCONJUGATION OVER D-ORBITAL PARTICIPATION. *JOURNAL OF THE AMERICAN CHEMICAL SOCIETY*, **112**(4), 1434–1445.
- SCRIVEN, L., & STERNLING, C. (1960). MARANGONI EFFECTS. *NATURE*, **187**(4733), 186–188. <https://doi.org/10.1038/187186a0>
- Solling, T. I., Andersson, M. P., Abdel-Azeim, S., El Zohry, A. M., & Beauchamp, J. L. (2024). Transition Metal Ions in Solution: Complexed Ion Speciation at the Air-Liquid Interface. *LANGMUIR*, **41**(3), 1577–1586. <https://doi.org/10.1021/acs.langmuir.4c03115>
- Solling, T. I., El Zohry, A. M., Ali, A. H. M. E., Baldelli, S., & Beauchamp, J. L. (2024). Tools to Investigate Structure at Hydrocarbon-Air Interfaces: Field-Induced Droplet Ionization Mass Spectrometry and Sum-Frequency Generation with Pulsed Lasers. *ENERGY & FUELS*, **38**(4), 2874–2882. <https://doi.org/10.1021/acs.energyfuels.3c04569>
- Solling, T., Olesen, K. B., Palm-Henriksen, G., Abdel-Azeim, S., & Skov, A. B. (2020). Crystalizing the interface—The first X-Ray structure of an oil/surfactant/brine transition layer. *JOURNAL OF PETROLEUM SCIENCE AND ENGINEERING*, **188**, 106953. <https://doi.org/10.1016/j.petrol.2020.106953>
- Yao, C., & Turecek, F. (2005). Hypervalent ammonium radicals. Competitive N-C and N-H bond dissociations in methyl ammonium and ethyl ammonium. *PHYSICAL CHEMISTRY CHEMICAL PHYSICS*, **7**(5), 912–920. <https://doi.org/10.1039/b414764b>
- Yin, Y., & Alivisatos, A. (2005). Colloidal nanocrystal synthesis and the organic-inorganic interface. *NATURE*, **437**(7059), 664–670. <https://doi.org/10.1038/nature04165>
- Zhdankin, V. V. (2016). Hypervalent Heterocycles. In E. F. V. Scriven & C. A. Ramsden (Eds.), *HETEROCYCLIC CHEMISTRY IN THE 21ST CENTURY: A TRIBUTE TO ALAN KATRITZKY* (Vol. **119**, pp. 57–79). <https://doi.org/10.1016/bs.aihch.2015.11.001>
- Zhou, J., Wang, Q., Cheng, G., Shen, W., Zare, R. N., & Sun, X. (2025). Charged Water Microdroplets Enable Dissociation of Surrounding Dioxygen. *JOURNAL OF THE AMERICAN CHEMICAL SOCIETY*. <https://doi.org/10.1021/jacs.4c12740>